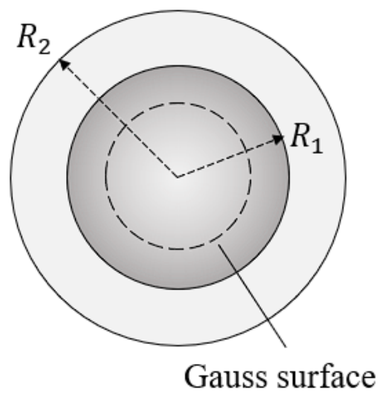
1. In a charged insulator sphere of radius R_1 and permittivity ε, the charge density is proportional to the distance to the center:



$$\rho = \beta r$$

where β is a constant. And the sphere is closely covered by a spherical conductor shell of outer radius R_2. By dividing the space into "thin spherical shells":

(a) Determine the electric field all over the space: including $r < R_1, R_1 < r < R_2, r > R_2$. (**Hint:** because the charge distribution is not uniform, the charge enclosed by a spherical gauss surface should be determined by making integral.)

(b) Determine the electrostatic potential at the center. (**Hint:** you must make integral from the infinity, where the potential is zero. Take care how to make integral for a piecewise function.)

(c) Determine the energy stored in the electric field in the insulator sphere.

[Solution] (a) The charge on a "thin shell": $\mathrm{d}q = \rho \times 4\pi r^2 \mathrm{d}r$

For $r < R_1$, the charge enclosed by a spherical gauss surface of radius r:

$$q_{\text{in}} = \int_0^r 4\rho\pi r^2 \mathrm{d}r = \int_0^r 4\beta\pi r^3 \mathrm{d}r = \beta\pi r^4$$

The electric field: $4\pi r^2 D = \beta\pi r^4$, $E = \beta r^2/4\varepsilon$.

For $R_1 < r < R_2$, in the conductor, E=0.

For $r > R_2$, $q_{\text{in}} = \int_0^{R_1} 4\rho\pi r^2 \mathrm{d}r = \beta\pi R_1^4$; the electric field: $E = \beta R_1^4/4r^2\varepsilon_0$

The direction: radian to the center.

(b) Take note that the minus sign can be removed by exchanging the upper and lower limit of the integral:

$$V = \int_{R_2}^{\infty} \frac{\beta R_1^4}{4r^2\varepsilon_0}\mathrm{d}r + \int_0^{R_1} \frac{\beta r^2}{4\varepsilon}\mathrm{d}r = \frac{\beta R_1^4}{4R_2\varepsilon_0} + \frac{\beta R_1^3}{12\varepsilon}$$

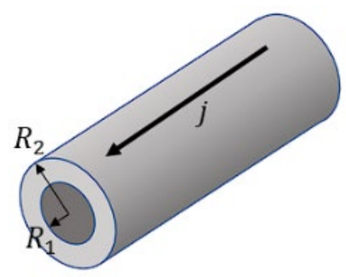
(c) The energy density of radius r and the energy on the "thin shell":

$$u = \frac{1}{2}\varepsilon E^2 = \frac{\beta^2 r^4}{32\varepsilon}, \mathrm{d}U = u \times 4\pi r^2 \mathrm{d}r = \frac{\pi\beta^2 r^6}{8\varepsilon}\mathrm{d}r$$

Total energy:

$$U = \int_0^{R_1} \frac{\pi\beta^2 r^6}{8\varepsilon}\mathrm{d}r = \frac{\pi\beta^2}{56\varepsilon}R_1^7$$

2. An infinitely long straight conductor tube, of which the inner radius is R_1 and the outer radius is R_2, carries a steady current along its axis, and the current density is j all over its cross area (横截面). The permeability of the tube is the same as vacuum. By selecting closed paths as circles perpendicular around the axis, determine the magnetic field distribution all over the space.



[Solution] For $r < R_1$, I_{in}=0, B=0.

For $R_1 < r < R_2$, $I_{\text{in}} = jA = \pi j(r^2 - R_1^2)$

$$2\pi r H = \frac{2\pi r}{\mu_0}B = \pi j(r^2 - R_1^2), B = \frac{\mu_0 j(r^2 - R_1^2)}{2r}$$

For $r > R_2$, $I_{\text{in}} = jA = \pi j(R_2^2 - R_1^2)$

$$B = \frac{\mu_0 j(R_2^2 - R_1^2)}{2r}$$

3. In the vicinity of position x=1, y=-1, z=0, the electric field at a certain moment is

$$\mathbf{E} = (2x - y)\alpha\mathbf{i} + 3\beta\mathbf{j} - yz\gamma\mathbf{k}$$

α, β, γ are all constants, and it is vacuum in this area.

(a) Determine the charge density here.

(b) Does a changing magnetic field exist here? If yes, determine its changing rate $\frac{\partial \mathbf{B}}{\partial t}$.

[Solution] (a) $\rho = \nabla \cdot \mathbf{D} = \varepsilon_0 \nabla \cdot \mathbf{E} = (2\alpha - y\gamma)\varepsilon_0$

At y=-1, $\rho = (2\alpha + \gamma)\varepsilon_0$

(b) Faraday's law:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (2x-y)\alpha & 3\beta & yz\gamma \end{vmatrix} = z\gamma\mathbf{i} + \alpha\mathbf{k}$$

So $\frac{\partial \mathbf{B}}{\partial t} = -z\gamma\mathbf{i} - \alpha\mathbf{k}$. There exists a changing magnetic field

4. (a) For an arbitrary vector field **X**, prove that the divergence of its curl is always zero:

$$\nabla \cdot (\nabla \times \mathbf{X}) = 0$$

And, if **X** is the magnetic vector potential, what law of magnetic field does it indicate?
(b) For an arbitrary scalar field Y, prove that the curl of its gradient is always zero:

$$\nabla \times \nabla Y = \mathbf{0}$$

And, if Y is the electrostatic potential, what law of electrostatic field does it indicate?
(**Hint**: for 2nd-order mixed partial derivative (二阶混合偏导), you may use $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ in a physical case.)

[Solution] (a)

$$\nabla \times \mathbf{X} = \left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}\right)\mathbf{i} + \left(\frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}\right)\mathbf{j} + \left(\frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)\mathbf{k}$$

$$\nabla \cdot (\nabla \times \mathbf{X}) = \frac{\partial}{\partial x}\left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)$$

$$= \frac{\partial^2 X_z}{\partial y \partial x} - \frac{\partial^2 X_y}{\partial z \partial x} + \frac{\partial^2 X_x}{\partial z \partial y} - \frac{\partial^2 X_z}{\partial x \partial y} + \frac{\partial^2 X_y}{\partial x \partial z} - \frac{\partial^2 X_x}{\partial y \partial z} = 0$$

If **X** is the magnetic vector potential, $\nabla \times \mathbf{X} = \nabla \times \mathbf{A}_{\mathrm{M}} = \mathbf{B}$, then $\nabla \cdot \mathbf{B} = 0$
This is the Gauss's law of magnetic field.
(b)

$$\nabla \times \nabla Y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial Y/\partial x & \partial Y/\partial y & \partial Y/\partial z \end{vmatrix}$$

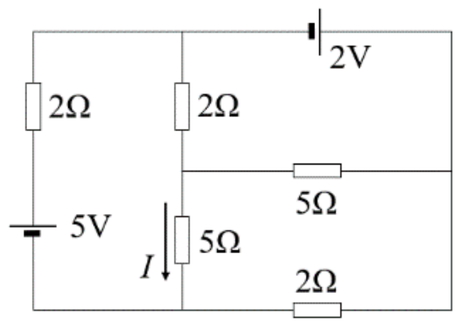
$$= \left(\frac{\partial^2 Y}{\partial y \partial z} - \frac{\partial^2 Y}{\partial z \partial y}\right)\mathbf{i} + \left(\frac{\partial^2 Y}{\partial z \partial x} - \frac{\partial^2 Y}{\partial x \partial z}\right)\mathbf{j} + \left(\frac{\partial^2 Y}{\partial x \partial y} - \frac{\partial^2 Y}{\partial y \partial x}\right)\mathbf{k} = \mathbf{0}$$

If Y is the electrostatic potential, $\nabla Y = -\mathbf{E}$, $\nabla \times \mathbf{E} = \mathbf{0}$.
The electric field is electrostatic field here. The integral form of the formula is

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = \int \mathbf{0} \cdot \mathrm{d}\mathbf{A} = 0$$

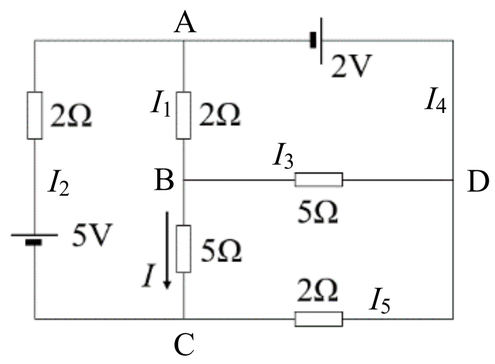
This is the loop theorem of electrostatic field, i.e. electrostatic field is conservative.

5. A circuit is shown on the right. Determine the current I indicated by the arrow. (**Attention:** it should be algebraic. If the result is opposite to the arrow, it is negative.)



[Solution] There are 6 branches in this circuit. Because all the components are determinate, we only need 6 equations.
For loops ABCA, ADBA and BCDB:

$$I_2 - 2I_1 + 5I = 5$$
$$5I_3 + 2I_1 = 2$$
$$5I_3 + 5I + 2I_5 = 0$$



For junctions A, C, D:

$$I_1 + I_2 = I_4 \;\rightarrow\; I_1 + I_2 - I_4 = 0$$
$$I_2 + I_5 = I \;\rightarrow\; I_2 + I_5 - I = 0$$
$$I_4 + I_5 = I_3 \;\rightarrow\; -I_3 + I_4 + I_5 = 0$$

The equations can be written as:

$$\begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 5 \\ 2 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 2 & 5 \\ 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

By transformation, it can be obtained:

$$\begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 5 & 5 \\ 2 & 0 & 5 & 0 & 0 & 0 & 2 \\ 0 & 0 & 5 & 0 & 2 & 5 & 0 \\ 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & 1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 5 & 0 & 0 & 5 & 7 \\ 0 & 0 & 5 & 0 & 2 & 5 & 0 \\ 0 & 0 & 0 & 1 & 2.8 & 5.5 & -1 \\ 0 & 0 & 0 & 0 & 3 & -1 & -7 \\ 0 & 0 & 0 & 0 & 0 & 149/30 & 34/15 \end{pmatrix}$$

According to the last row, there is

$$\frac{149}{30} I = \frac{34}{15} \quad \rightarrow \quad I = 0.456(\mathrm{A})$$